

1 Training

DP0 Sparse MoE 3
DP x 4 PP

DP1

DP2



2 Detect + Repair (R1)

Failure Event

Repair Controller
Safe-Point Guard

- recovering
- block opt
- discard iter.

YES

3 Guarded Policy (R2)

Score

$\Phi'(t)$

hybrid vs restart

HYBRID?

NO

4a Hybrid Recovery

Peers (t)

Repl.
 r'

Ckpt

Two-phase

weights

resume
barrier

opt

Reintegration

RECOVER

REPAIRED

BARRIER

HEALTHY



4b Checkpoint Restart

Checkpoint

Reload

Replay

Resume

5 Logs & Metrics (R3)



Audit trail

policy + reason

state trace

latency

quality

